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Observed and Simulated Solute Transport Under Varying Water Regimes: II. 2,6-Difluorobenzoic Acid and Dicamba

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ABSTRACT

Significant interest in the fate of agrichemicals in soils has prompted the development of several transport simulation models. Our primary objective was to evaluate the simulation model LEACHM for predicting the fate of dicamba (3,6-dichloro-2-methoxybenzoic acid) and a nonreactive tracer, 2,6-difluorobenzoic acid (2,6-DFBA) in fallow and cropped (barley, *Hordeum vulgare* L.) systems under different water application levels. A field study (1992) was conducted using in situ soil columns on a Borollic Calciorthid (Brocko silt loam, Gallatin Co., MT). Dicamba (^{14}C-labeled) and 2,6-DFBA were surface applied at rates of 0.26 and 112 kg ha^{-1}, respectively. Solute concentrations were measured at three depths (0.36, 0.66, and 0.96 m) using porous cup lysimeters for 70 d following chemical application. Time-moment analysis of observed solute breakthrough curves (BTCs) generally showed increasing travel times with increasing soil depth and decreasing water application. Dicamba transport was similar to 2,6-DFBA, with the exception that 40 to 60% of applied dicamba was degraded during transport. The distribution of ^{14}C remaining in the soil columns showed that the primary degradate of dicamba, 3,6-dichlorosalicylic acid (DCSA), was confined primarily to surface samples (0–0.2 m) while dicamba was found only at lower depths. This is consistent with a much higher sorption coefficient (K_{oc}) determined for DCSA relative to dicamba. Comparison of observed and predicted (LEACHM) solute BTCs suggested that preferential solute transport occurred especially under high and medium water regimes. Finally, data from three field seasons at the same site suggest that the timing of chemical application relative to initial soil water content and plant stage, the presence of root channels, and temporal changes in soil hydraulic properties in the absence of tillage may significantly affect the degree of preferential flow and subsequent agreement between predicted and observed BTCs.

Pesticide leaching and subsequent contamination of groundwaters from agricultural production is a significant national concern. Recent monitoring programs have detected more than 70 pesticides in groundwaters of 38 states (Ritter, 1990; Parsons and Witt, 1988). In one survey, 17 pesticides were detected at concentrations above health advisory limits. DeLuca et al. (1989) and Clark (1990) conducted well water monitoring surveys in agricultural regions in Montana and documented that several pesticides (including aldicarb, atrazine, 2,4-D, dicamba, MCPA, picloram, and simazine) have migrated into shallow groundwaters, presumably through normal agricultural management practices.

Dicamba (3,6-dichloro-2-methoxybenzoic acid) is commonly used in Montana to control broadleaf weeds in small grain production systems and has been identified as a groundwater contaminant (DeLuca et al., 1989). Dicamba is anionic at most soil pH values (pK_a = 1.95) and is highly soluble in water (6.5 × 10^3 mg L^{-1}) (Pesticide Manual, 9th ed., 1991). Consequently, dicamba has a low sorption coefficient (Burnside and Lavy, 1966; Grover and Smith, 1974) and is highly mobile in most soils (Friesen, 1965; Grover, 1977; Scifres and Allen, 1973). Soil environmental conditions play a significant role in the fate and mobility of dicamba. Degradation rates of dicamba in surface soils are generally rapid with half-lives ranging from 13.5 to 45 d (Comfort et al., 1992; Krueger et al., 1991; Smith, 1974; Smith and Cullimore, 1974). The primary degradate of dicamba is DCSA, which is more persistent and less mobile than the parent compound (Smith, 1973a; Smith, 1974; Comfort et al., 1992). Consequently, the potential for dicamba to leach out of the rooting zone increases dramatically if soil conditions limit its degradation rate. In a laboratory column and incubation study, Comfort et al. (1992) showed that dicamba transport and subsequent leaching losses could be reduced substantially by delaying water application and providing time for dicamba to degrade to the less mobile DCSA.

Because of regional concerns about the potential mobility of broadleaf herbicides, we have been studying the fate and mobility of 2,4-D and dicamba under different soil water regimes, in both laboratory (Comfort et al., 1992; Veeh et al., 1996) and field experiments. The specific objectives of this study were to: (i) monitor the transport and fate of dicamba and a nonsorbing tracer, 2,6-DFBA, under fallow and cropped (barley) conditions and different soil water regimes, and (ii) evaluate the suitability of the computer simulation model LEACHM (Wagenet and Hutson, 1989) for predicting the fate of these compounds under field conditions, given site-specific soil and climate input parameters. This is the second of two manuscripts (see Pearson et al., 1996) which compare observed and simulated solute transport over a range of applied water and evapotranspirative demands.

MATERIALS AND METHODS

Field Design

An in situ column experiment was conducted in 1992 on a Brocko silt loam (Borillic Calciorthid) soil to study the transport of ^{14}C-labeled dicamba and a nonsorbing tracer, 2,6-DFBA. Eighteen of the 24 PVC columns (0.20 m diam., 1.22 m depth) described in the companion paper (Pearson et al., 1996) were used to establish three water regimes (high, medium, and low) on continuous crop and continuous fallow treatments, with three replications per treatment.

The crop treatments were seeded to barley (cv. Klages) on 1 May 1992, and thinned to eight plants per column (approximately one plant per 38 cm^2) on 12 May to match the plant

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Abbreviations: DCSA, 3,6-dichlorosalicylic acid; DFBA, 2,6-difluorobenzoic acid; PET, potential evapotranspiration; BTCs, breakthrough curves; ET, evapotranspiration.

density obtained outside the columns. Solutions (100 mL) containing ^{14}C-ring labeled dicamba and 2,6-DFBA were uniformly applied dropwise to the surface of each soil column on 18 June 1992 (48 d after crop treatments were seeded). Dicamba was applied at a normal field application rate of 0.26 kg ha^{-1} (16 uCi ^{14}C per column) and 2,6-DFBA at 112 kg ha^{-1}. Dicamba was applied only to the high and medium water regime columns. Water was applied eight times (approximately once per week) during 70 d following chemical application, resulting in a total water application (irrigation plus precipitation) of 406, 344, and 278 mm to the high, medium, and low water regimes, respectively. For comparison, the potential evapotranspiration (PET) measured on site using pan evaporation was 440 mm during this same 70-d period. Of the total water applied, 75 mm was precipitation. Soil solution samples were vacuum extracted 27 times during the experiment as described in Pearson et al. (1996). Volumetric soil water contents (θ_v, m^3 m^{-3}) and bare-soil evaporation were monitored as described in Pearson et al. (1996).

Sample Analysis

All soil solution samples were analyzed for total ^{14}C using a Packard 2200 CA liquid scintillation analyzer. Selected samples were analyzed for dicamba and degradates using an HPLC equipped with a Lichrosorb RP-18 column (EM Science, Gibbstown, NJ) and Beckman model 171 radioisotope detector (Beckman Instruments Inc., Fullerton, CA). In all cases, soluble ^{14}C was identified as dicamba with no detectable levels of the principal degradate, DCSA. All lysimeter samples were analyzed for 2,6-DFBA by ion chromatography using a Dionex AS4A column (Pearson et al., 1992).

On 27 Aug. 1992 all soil columns were removed and later sectioned into 0.1-m depth increments. The soil was dried at 37°C for 48 h, then ground to pass through a 2-mm sieve. Triplicate 1.0-g subsamples from each 0.10-m depth increment were oxidized (Model OX300, R.J. Harvey Instrument Corp., Hillsdale, NJ) and analyzed for total ^{14}C using liquid scintillation analysis. Selected surface and subsurface samples (containing the highest concentration of total ^{14}C as a function of depth) were solvent extracted (Smith and Muir, 1980) to determine the amount of ^{14}C present as dicamba or degradates. Fifty grams of soil were extracted with 100 mL of 70% acetonitrile, 27% H_2O, and 3% glacial acetic acid and shaken for 12 h. After filtration, the acidified samples were extracted into methylene chloride (CH_2Cl_2) and analyzed using HPLC-radioisotope detection (described previously). Percent recoveries of spiked soil samples for dicamba and DCSA using this methodology were 86 and 47%, respectively. Efforts to improve the extraction recovery of DCSA with this extractant were not successful.

Concentrations of ^{14}C-dicamba and 2,6-DFBA were plotted as functions of time for each lysimeter depth (0.36, 0.66, and 0.96 m) to establish breakthrough curves (BTCs) for each chemical. Moment analysis (Skopp, 1984) of complete BTCs (primarily the 0.36 and 0.66 m depths) was used to estimate mean travel times for solute center of mass (d), dispersion coefficients (D, cm^2 d^{-1}), and average pore water velocities (cm d^{-1}). Percent recoveries for each chemical were calculated based on the fraction of applied chemical mass measured in the BTC at each lysimeter depth.

LEACHM Simulations

Predicted BTCs for 2,6-DFBA and dicamba were generated using the LEACHP subroutine in LEACHM (version 2, Wagenet and Hutson, 1989) with independent measurements or estimates of chemical, soil, climate, and plant parameters as described in Pearson et al. (1996). The sorption coefficient (K_{oc}) for 2,6-DFBA was assumed to be 0. Batch adsorption studies were used to obtain independent estimates of K_{oc} for dicamba and its principal degradate, DCSA. Duplicate soil suspensions (10 g soil from 0–15 cm: 10 mL H_2O) were spiked with ^{14}C ring-labeled dicamba or DCSA at concentrations ranging from 0.24 to 2.25 mg L^{-1} (4.56 $\times 10^{-4}$ to 4.56 $\times$ 10^{-2} uCi ^{14}C mL^{-1}) and 0.01 to 0.70 mg L^{-1} (4.81 $\times$ 10^{-4} to 4.81 $\times$ 10^{-2} uCi ^{14}C mL^{-1}) for dicamba and DCSA, respectively. These concentration ranges bracketed the levels of dicamba and DCSA measured in soil solution samples during the field experiment. The soil suspensions were shaken for 48 h, then centrifuged at 2000 $\times$ g for 15 min. A 1-mL aliquot of the supernatant solution was used to determine ^{14}C activity by liquid scintillation analysis, and the amount of DCSA or dicamba sorbed was determined by difference between total added and that remaining in the supernatant. The amount of dicamba sorbed was not statistically different than 0 (i.e., K_{oc} = 0) over the range of dicamba levels investigated, while the sorption coefficient (K_{oc}) for DCSA was 504 L kg^{-1}. An average molecular diffusion coefficient (D_o) of 6 $\times$ 10^{-5} m^2 d^{-1} was used for 2,6-DFBA, dicamba, and DCSA. As mentioned in Pearson et al. (1996), LEACHM predictions of solute concentrations were insensitive to D_o over several orders of magnitude. First-order degradation rate constants for dicamba and DCSA for the soil surface horizons were estimated from previous batch studies in our laboratory (Comfort et al., 1992) and from published values (Smith, 1974; Krueger et al., 1991). Values of 0.051 d^{-1} ($t_{1/2}$ = 13.5 d) for dicamba and 0.017 ($t_{1/2}$ = 40 d) for DCSA were used for the soil surface horizon (0–30 cm). Degradation rates of herbicides often decrease with soil depth as a result of reduced microbial activity (Moorman and Harper, 1989; Pothuluri et al., 1990; Veeh et al., 1996). Half-lives for soil depths >30 cm were estimated using an exponential decay function as outlined by Jury et al. (1987):

$$t_{1/2}(Z) = t_{1/2}(\text{surface})e^{\gamma(Z-L)} \quad [1]$$

where γ is a depth constant (1.5 m^{-1}), Z is soil depth, and L is the depth of the surface layer. Although these are estimated values, we have found that this function provides a reasonable description of the variation in 2,4-D degradation as a function of soil depth (Veeh et al., 1996) and of the amount of organic C (which often relates to microbial activity) as a function of soil depth (Wilson et al., 1993).

RESULTS AND DISCUSSION

Observed Dicamba and 2,6-DFBA Breakthrough Curves

As expected, travel times required to detect solute fronts for dicamba and 2,6-DFBA increased with increasing soil depth and generally increased with decreasing water application (Fig. 1 through 4). Under barley, little difference in travel times was observed for solute BTCs under high and medium water regimes (Fig. 2 and 4). For example, solute fronts of 2,6-DFBA and dicamba were observed within 5 d at 0.36 m and within 22 d at 0.66 m for both the high and medium water regimes (Fig. 2 and 4). Unlike PFBA transport reported for the 1991 field season (Pearson et al., 1996), solute travel times were significantly shorter under crop compared to fallow conditions. For example, under fallow treatments, 2,6-DFBA solute fronts were detected within 12 d at

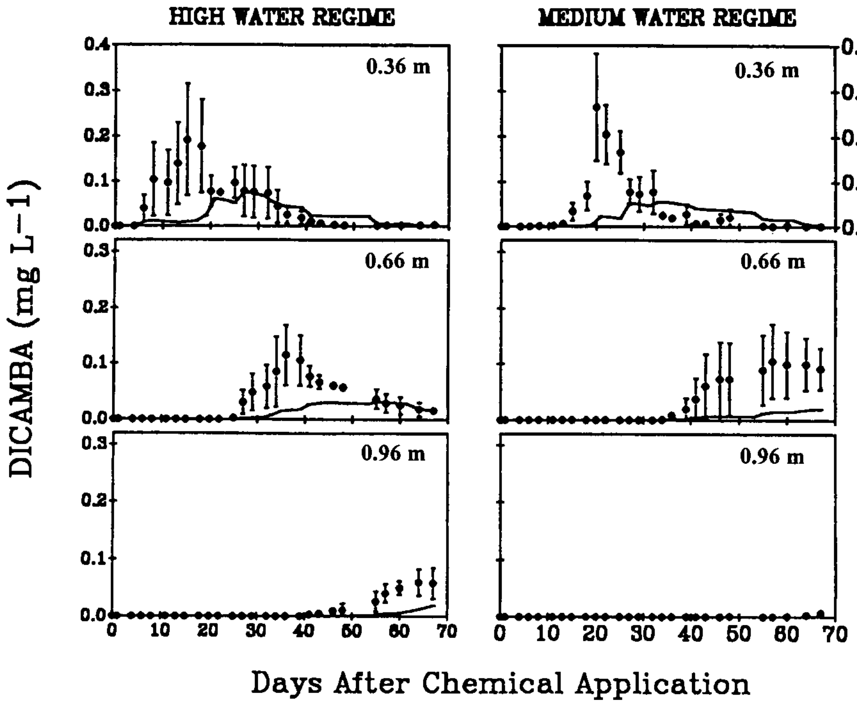

Fig. 1. Observed (●) and simulated (−) dicamba concentrations at 0.36, 0.66, and 0.96 m depths under high and medium water regimes for fallow treatments. Vertical bars on observed data indicate standard errors (n = 3); where absent, bars fall within symbols.

0.36 m and 36 d at 0.66 m for the medium water regime (Fig. 3). Under barley, solute fronts were detected within 5 d at 0.36 m and 22 d at 0.66 m (Fig. 4). This trend was also observed in the low water regime where 2,6-DFBA fronts at 0.36 m exhibited shorter travel times under barley (15 d) compared to fallow (22 d) conditions (Fig. 3 and 4).

Trends in mean travel times (d) obtained from moment

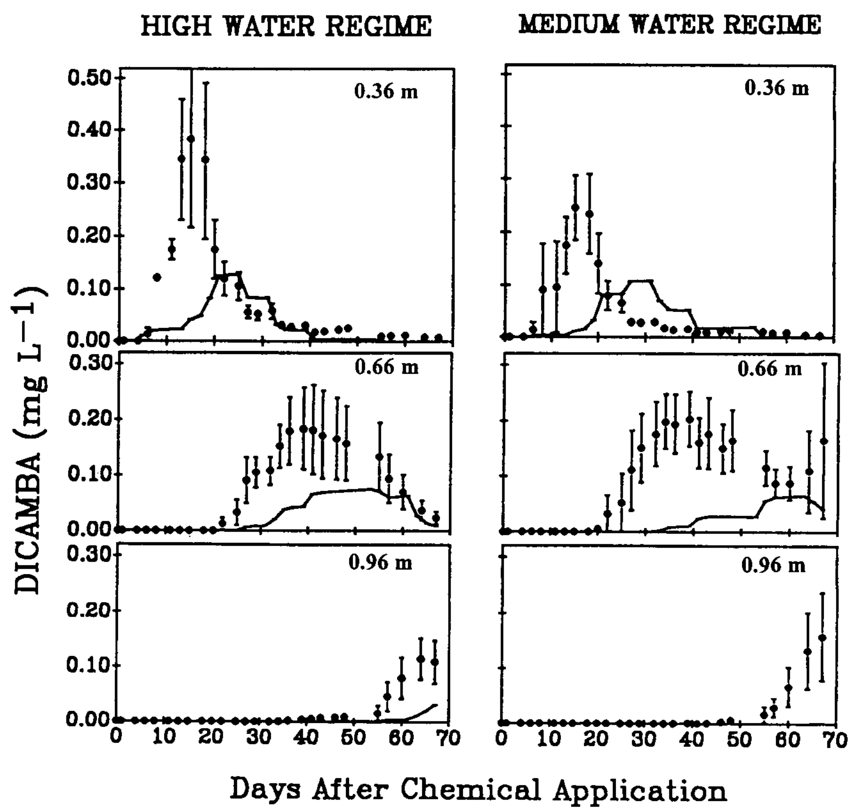

Fig. 2. Observed (●) and simulated (−) dicamba concentrations at 0.36, 0.66, and 0.96 m depths under high and medium water regimes for crop treatments. Vertical bars on observed data indicate standard errors (n = 3); where absent, bars fall within symbols.

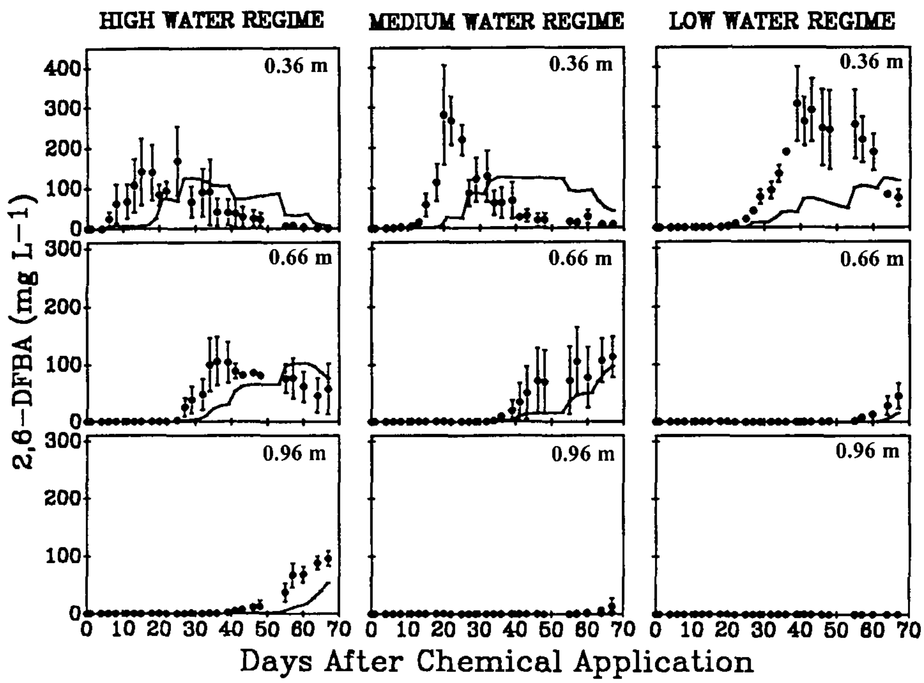

Fig. 3. Observed (●) and simulated (–) 2,6-DFBA concentrations at 0.36, 0.66, and 0.96 m depths under high, medium, and low water regimes for fallow treatments. Vertical bars on observed data indicate standard errors (n = 3); where absent, bars fall within symbols.

analysis (Skopp, 1984) were consistent with trends reported above based on time required to observe solute fronts. However, results from moment analysis are presented (Table 1) only for those solute BTCs where solute concentrations returned to near baseline levels by 70 d after chemical application (last day of lysimeter sampling). Dicamba and 2,6-DFBA BTCs consistently showed longer mean travel times with decreasing water application in the fallow treatment (Table 1). However, in the cropped treatments, mean travel times were essentially identical at high and medium water regimes. Solute BTCs generally showed shorter mean travel times at a given depth and a given water regime under crop compared to fallow conditions (Table 1). Finally, average (n = 3) mean travel times (d) were slightly greater for 2,6-DFBA than dicamba for all depths and water regime-cropping treatments. The measured dicamba K_{oc} value for the surface (0–15 cm) layer was essentially 0

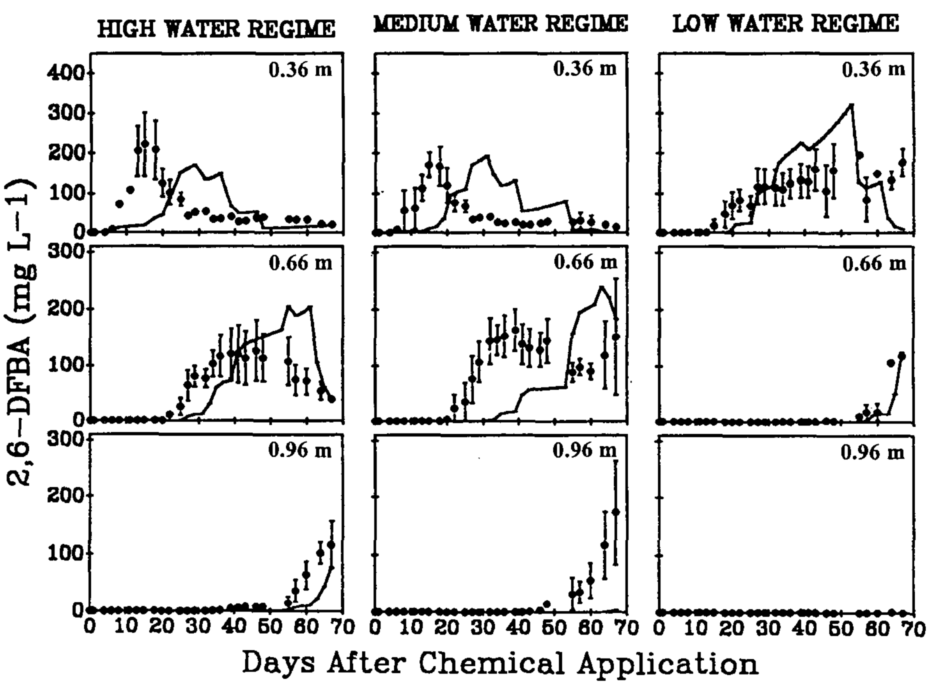

Fig. 4. Observed (●) and simulated (–) 2,6-DFBA concentrations at 0.36, 0.66, and 0.96 m depths under high, medium, and low water regimes for crop treatments. Vertical bars on observed data indicate standard errors (n = 3); where absent, bars fall within symbols.

Table 1. Moment analysis results of selected† observed and predicted (LEACHM) dicamba and 2,6-DFBA breakthrough curves.

			Mean travel time							
			Observed		Predicted		D§		Recovery	
Soil depth	Water regime	Treatment	Dicamba	2,6-DFBA	Dicamba	2,6-DFBA	Dicamba	2,6-DFBA	Dicamba	2,6-DFBA
m			d				$cm^2\ d^{-1}$		%	
0.36	High	Fallow	21.4 (4.5)¶	23.0 (4.8)	31.9	37.5	3.4	3.6	78.1	131.7
		Crop	21.0 (1.0)	26.9 (2.6)	24.2	27.3	10.1	8.1	116.7	139.3
	Medium	Fallow	25.9 (1.9)	29.4 (3.8)	39.1	44.0	2.1	2.7	90.6	130.6
		Crop	20.5 (2.8)	26.2 (3.8)	30.6	34.3	8.9	7.6	67.7	94.6
	Low	Fallow	ND‡	46.6 (0.9)	ND	51.5	ND	0.6	ND	120.1
		Crop	ND	46.1 (1.8)	ND	44.5	ND	1.2	ND	91.4
0.66	High	Fallow	43.6 (4.0)	46.2 (3.7)	49.9	52.6	1.7	1.8	42.4	100.1
		Crop	41.4 (3.5)	43.4 (2.6)	48.6	51.0	2.8	3.2	97.6	138.0
	Medium	Crop	43.0 (3.9)	44.7 (2.9)	54.4	56.4	3.0	3.1	103.5	153.9

† Values from truncated breakthrough curves do not represent true moments and are not presented.
‡ Not determined - dicamba was not applied to low water regime treatments.
§ D = Dispersion coefficient.
¶ Standard errors in parentheses.

for this soil. It has been shown in other studies (Bowman, 1984; Bowman and Gibbens, 1992) that the K_{oc} value for 2,6-DFBA is also 0. The primary cause of shorter mean travel times for dicamba is microbial degradation during transport, which causes the concentration of dicamba to decrease as a function of time.

We originally hypothesized that the transport of dicamba and 2,6-DFBA under barley would lag behind transport under fallow, consistent with observations of PFBA transport from the previous field season (Pearson et al., 1996). However, despite higher evapotranspiration (ET) in crop vs. fallow treatments, dicamba and 2,6-DFBA transport was generally faster in the cropped columns. Results from studies conducted by Meek et al. (1992) and Zins et al. (1991) have shown that the presence of alfalfa roots increases infiltration rates through soils. Beven and Germann (1982) suggested that preferential flow pathways are associated with live and decayed roots. Gish and Jury (1983) reported dispersivities of Cl transport through a sandy loam soil as influenced by dead wheat root systems. The dispersivities they observed were characteristic of dispersivities obtained in a "porous medium with significant flow occurring in large pores". Moment analysis of our data showed that dispersion coefficients at 0.36 m for both 2,6-DFBA and dicamba were roughly three times higher in cropped than fallow columns (Table 1) for both the high and medium water regimes. One explanation for shorter travel times and greater solute dispersion in crop treatments is a greater heterogeneity of pore water velocities resulting from root channels. Although this trend was not observed for PFBA in the 1991 field season (Pearson et al., 1996), chemical application was made 18 d after seeding in 1991 compared to 48 d after seeding in 1992. Consequently, a more developed root system would have been established at the time of chemical application in 1992 compared to 1991.

Moment analysis was also used to estimate the percent of applied solute measured in individual solute BTCs (Table 1). Percent recoveries ranged from 91 to 150% for 2,6-DFBA and from 42 to 117% for dicamba over all treatments where analysis of complete BTCs was possible. The fact that recoveries for dicamba were roughly 20 to 40% lower (average of 33%) than recoveries for 2,6-DFBA indicates that degradation of dicamba was occurring during transport. As observed for PFBA in the 1991 field season (Pearson et al., 1996), no significant trends in percent recoveries of 2,6-DFBA were noted between crop and fallow treatments (Table 1), suggesting that this tracer was not subject to significant plant uptake during transport.

Observed vs. Simulated Solute Breakthrough Curves

Predicted (LEACHM) dicamba and 2,6-DFBA BTCs were significantly slower than observed BTCs for the majority of treatment-water regime combinations (Fig. 1 through 4). Predicted mean travel times for dicamba BTCs averaged over all depths and water regimes were between 27 and 38% higher than observed for crop and fallow treatments, respectively (Table 1). Likewise, predicted mean travel times for 2,6-DFBA BTCs were between 14 and 34% higher than observed for crop and fallow treatments. Predicted BTCs for the nonreactive tracer, 2,6-DFBA, were based predominantly on input parameters that influence water movement and are essentially identical to predicted BTCs of dicamba (K_{oc} for both solutes was set to 0), with the exception that dicamba is degraded to DCSA during transport. Consequently, lack of agreement between observed and predicted solute BTCs was due either to transport (i.e., physical rather than chemical) processes not described by the model, or to inadequate input data for soil physical properties.

However, adequate agreement between model predictions and observed data was obtained for evaporation under fallow conditions (Fig. 5) and volumetric water contents at 0.4, 0.6, and 1.0 m soil depths (data not shown). The only major disagreement between observed and predicted θ_v in 1992 was at 1.0 m for the low water regime under cropped conditions, where LEACHM overestimated the measured θ_v. As discussed in Pearson et al. (1996), this may be due to an inadequate description of root distribution and subsequent depth-dependent root water uptake.

Several factors may be responsible for the greater

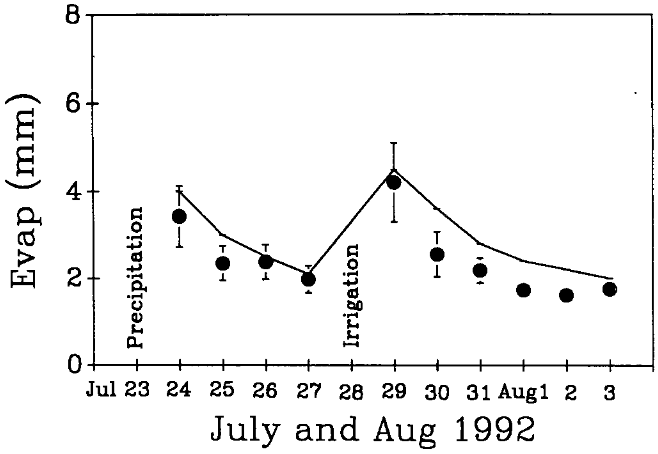

Fig. 5. Observed (●) and simulated (–) daily surface evaporation from bare soil over two wetting-drying cycles.

disagreement between observed and predicted BTCs in the current study compared to studies conducted during the previous two field seasons (Comfort et al., 1993; Pearson et al., 1996). First, actual θ_v values immediately following chemical application were significantly higher (to depths of 1 m) in the current study (ranged from 0.30–0.34) compared to the 1991 field season (ranged from 0.22–0.25) because of the application of two irrigation treatments between 0 and 5 d after chemical application. Higher θ_v's immediately following chemical application would have contributed to increased potential for preferential flow compared to the previous field season. Agreement between observed and predicted BTCs would be expected to improve under the low water regime where preferential solute flow would be less important. This is in fact what we observed for 2,6-DFBA transport at the low water regime (Fig. 3 and 4), where observed and predicted mean travel times agreed within 7% (Table 1).

Secondly, the current study (1992 field season) was conducted on in situ columns that had not been recently disturbed by surface tillage (the fallow columns had been undisturbed for 2 yrs and the crop columns had been undisturbed for 1 yr). Consequently, temporal changes in soil physical properties such as soil aggregation and biopore formation (Hamblin, 1982; Hamblin and Tennant, 1981; Beven and Germann, 1982) may have contributed to the lack of agreement between observed and predicted BTCs in the current study. Such changes would be especially important in surface horizons (0–0.2 m) and would not have been accurately represented using input data based on adjacent intact soil cores. Increases in macroporosity combined with higher values of θ_v immediately following chemical application may have contributed to preferential flow, which would not have been described by LEACHM.

Finally, chemical application occurred 48 d after seeding in the current study compared to 18 d after seeding in 1991. Consequently, in the crop treatments, a greater number of active root channels would have been present immediately following chemical application in 1992. The presence of old and active root channels and the development of greater macroporosity in the absence of tillage combined with higher θ_v's immediately following chemical application may have resulted in greater preferential solute transport in 1992 compared to 1991.

As mentioned, we expected the transport of dicamba and 2,6-DFBA to be slower under barley compared to fallow conditions. However, both the observed and predicted mean travel times of 2,6-DFBA and dicamba were consistently shorter under barley, especially at 0.36 m (Table 1). Evaluation of LEACHM output indicated that, despite greater cumulative ET under cropped conditions, lower evaporation from the soil surface allowed more applied water to move into the profile where root uptake occured. For example, between chemical application and the mean travel time required for 2,6-DFBA to reach 0.36 m, LEACHM predicted that approximately 72, 71, and 66 mm more water evaporated for the high, medium, and low water regimes under fallow compared to crop treatments. Predicted root water uptake was distributed within the 0.05 to 0.8 m zone; consequently, a greater fraction of applied water was predicted to move past the 0.36 and 0.66 m depths under cropped vs. fallow conditions. Although a smaller fraction of applied water was predicted to evaporate under crop conditions, the predicted drainage component was lower for crop treatments because of root-water uptake occurring in the soil profile. The net effect of these processes resulted in predicted solute BTCs that were significantly faster at 0.36 and 0.66 m under cropped conditions, but slightly slower at 0.96 m (Table 1, Fig. 3 and 4). In the previous field season (Pearson et al., 1996), these trends were not observed because chemical application and subsequent transport to the 0.36 and 0.66 m depths occurred prior to the occurrence of significant root-water uptake.

Dicamba and DCSA Concentrations in the Soil Profile

At 70 d after chemical application, the majority of soil columns showed a bimodal distribution of total ^{14}C, with broad peaks in the 0 to 0.3 m and 0.8 to 1.2 m zones (Fig. 6). Total recoveries calculated as a fraction of applied ^{14}C ranged from 47 to 67% (Table 2), indicating that some of the applied ^{14}C-dicamba was completely mineralized to CO_2. Selected soil samples having the highest total ^{14}C contents were solvent extracted and analyzed for dicamba and the principal degradate, DCSA. The percent of ^{14}C extracted with acetonitrile ranged from 2 to 5% for surface samples and 50 to 130% for deep samples (Table 3). Dicamba was consistently identified as the predominant ^{14}C constituent at lower depths, consistent with the position of lysimeter measured dicamba concentrations at the time the columns were taken from the field and sectioned (Fig. 1 and 2). Roughly 90% of the extracted ^{14}C from deep samples was identified as dicamba while the remaining 10% was identified as DCSA (Table 3).

Solvent extraction recoveries of total ^{14}C were extremely low in surface samples (2–5%) and the only ^{14}C-labeled constituent identified in these extracts was DCSA. This observation is consistent with the fact that dicamba will degrade to DCSA fairly rapidly in surface

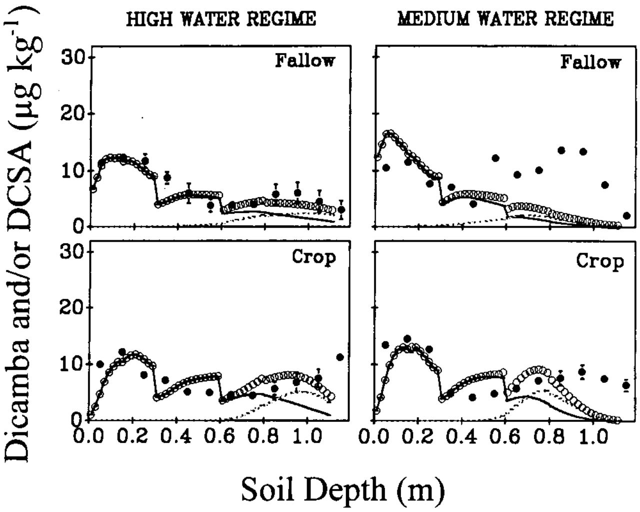

Fig. 6. Observed and simulated concentrations of dicamba and DCSA remaining in the soil profile at termination of field experiments under fallow and cropped conditions (○ = observed residual total ^{14}C, ● = simulated total dicamba plus DCSA, - - - = simulated dicamba, — = simulated DCSA). Vertical bars on observed data indicate standard errors (n = 3); where absent, bars fall within symbols.

horizons ($t_{1/2}$ = 13 d), leaving a greater fraction of the less mobile DCSA (measured K_{oc} value for DCSA = 504 L kg^{-1} compared to 0 for dicamba). Moreover, once DCSA is formed, its degradation rate ($t_{1/2}$ > 40 d) is considerably slower than dicamba. The nonextractable portion of the total ^{14}C in surface samples was probably because of the formation of irreversibly bound DCSA residues, although we cannot rule out the possibility of other nonextractable metabolites. The difference in extraction recoveries of ^{14}C in surface (DCSA) compared to deep samples (dicamba) is also consistent with the extraction recoveries measured on samples that were spiked with known masses (but not aged) of ^{14}C-labeled DCSA (47% recovery) and dicamba (86% recovery).

Predicted (LEACHM) soil concentrations of dicamba plus DCSA (i.e., total residual ^{14}C expressed as µg kg^{-1}) follow trends similar to the observed total residual ^{14}C (Fig. 6). LEACHM predictions show that essentially 100% of the parent dicamba is absent from the 0 to 0.4 m zone, while predicted DCSA concentrations are very close to observed total residual concentrations. This is consistent with the solvent extraction data showing that surface samples contained no detectable dicamba, and were dominated by DCSA. However, at depths >0.4 m, LEACHM predictions show less total residual dicamba plus DCSA than observed. The fact that predicted degradation rates were faster than observed is also demonstrated by comparing the predicted vs. observed recovery of total ^{14}C in the soil profile (i.e., amount of chemical left in the profile vs. amount applied) (Table 2). The predicted rate of dicamba and DCSA degradation is a function of the first-order rate constants (i.e., half-lives) used as input data to LEACHM. It is quite probable that actual degradation rates in the field were different than those estimated from laboratory experiments because of temporal variation in soil temperature (Comfort et al., 1992) and soil moisture (Smith, 1973b). Furthermore, it appears that actual degradation rates at lower soil depths were slower than those estimated using Eq. [1].

Table 2. Observed and predicted (LEACHM) ^{14}C recoveries in in situ soil columns (in 1.1 m) as a percent of total ^{14}C applied as dicamba.

Treatment	Water regime	% ^{14}C Recovered Observed	Predicted
Fallow	High	45.2 (5.7)†	38.6
	Medium	60.1 (3.2)	36.7
Crop	High	44.3 (4.0)	44.7
	Medium	61.0 (8.3)	43.0

† Standard errors in parentheses.

SUMMARY AND CONCLUSIONS

Observed dicamba BTCs show that it is mobile and will move significantly under irrigation. Dicamba behaves much like a nonsorbing tracer with the exception of any degradation that might occur. Consequently, the amount of dicamba that moves out of the root zone and potentially into shallow groundwaters is very sensitive to its residence time in the biologically active surface zone. Once dicamba degrades to DCSA, mobility is very low even under irrigation. Management practices that delay irrigation after dicamba application will significantly reduce the potential for transport of dicamba out of the root zone. These observations are consistent with those made by Comfort et al. (1992) in the laboratory.

Model predictions in this study were based on independently measured or estimated input parameters. Our intent was to compare a noncalibrated simulation of solute BTCs with observed data to assess LEACHM perfor-

Table 3. Characterization of ^{14}C fractions in selected soil samples (70 d after chemical application) using total oxidation, solvent extraction, and HPLC-radioisotope detection.

Treatment	Water regime	Soil depth	^{14}C Soil	^{14}C Extracted	^{14}C Extracted	Extracted ^{14}C† Dicamba	Extracted ^{14}C† DCSA
		m	dpm g^{-1}	dpm g^{-1}	%	%	%
Fallow	High	0.1-0.2	596.9	20.1	3.4	0.0	100.0
		0.9-1.0	356.0	479.7	134.7	88.5	11.5
	Medium	0.1-0.2	518.0	16.1	3.1	0.0	100.0
		0.8-0.9	611.0	432.2	70.7	87.8	12.2
Crop	High	0.1-0.2	571.3	27.5	4.8	0.0	100.0
		0.9-1.0	409.2	221.5	54.1	90.2	9.8
	Medium	0.1-0.2	614.8	14.4	2.3	0.0	100.0
		0.9-1.0	443.8	373.3	84.1	90.2	9.8

† Percent of total ^{14}C extracted as dicamba or 3,6-dichlorosalicyclic acid (DSCA). No other ^{14}C degradates identified.

mance under field conditions. LEACHM predicted mean travel times were significantly greater (i.e., slower transport) than observed for both dicamba and the nonreactive tracer 2,6-DFBA. Probable causes may include temporal changes in soil hydraulic properties not reflected in the measured set of input parameters and the inability of LEACHM to describe macropore-induced preferential flow. Comparison of field transport data over a 3-yr period at the same site show that the level of agreement between observed solute transport data and predicted (LEACHM) BTCs varies dramatically as a function of tillage history, the timing of chemical application relative to water application and crop stage, and the presence of active root channels.

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